**Learning Objectives**

After studying this chapter, you should be able to do the following:

1. List six different property classifications of materials that determine their applicability.
2. Cite the four components that are involved in the design, production, and utilization of materials, and briefly describe the interrelationships between these components.
3. Cite three criteria that are important in the materials selection process.
4. (a) List the three primary classifications of solid materials, and then cite the distinctive chemical feature of each.
   (b) Note the four types of advanced materials and, for each, its distinctive feature(s).
5. (a) Briefly define *smart material/system*.
   (b) Briefly explain the concept of *nanotechnology* as it applies to materials.

## 1.1 HISTORICAL PERSPECTIVE

Please take a few moments and reflect on what your life would be like without all of the materials that exist in our modern world. Believe it or not, without these materials we wouldn't have automobiles, cell phones, the internet, airplanes, nice homes and their furnishings, stylish clothes, nutritious (also "junk") food, refrigerators, televisions, computers . . . (and the list goes on). Virtually every segment of our everyday lives is influenced to one degree or another by materials. Without them our existence would be much like that of our Stone Age ancestors.

Historically, the development and advancement of societies have been intimately tied to the members' ability to produce and manipulate materials to fill their needs. In fact, early civilizations have been designated by the level of their materials development (Stone Age, Bronze Age, Iron Age).[1]

The earliest humans had access to only a very limited number of materials, those that occur naturally: stone, wood, clay, skins, and so on. With time, they discovered techniques for producing materials that had properties superior to those of the natural ones; these new materials included pottery and various metals. Furthermore, it was discovered that the properties of a material could be altered by heat treatments and by the addition of other substances. At this point, materials utilization was totally a selection process that involved deciding from a given, rather limited set of materials, the one best suited for an application by virtue of its characteristics. It was not until relatively recent times that scientists came to understand the relationships between the structural elements of materials and their properties. This knowledge, acquired over approximately the past 100 years, has empowered them to fashion, to a large degree, the characteristics of materials. Thus, tens of thousands of different materials have evolved with rather specialized characteristics that meet the needs of our modern and complex society, including metals, plastics, glasses, and fibers.

The development of many technologies that make our existence so comfortable has been intimately associated with the accessibility of suitable materials. An advancement in the understanding of a material type is often the forerunner to the stepwise progression of a technology. For example, automobiles would not have been possible without the availability of inexpensive steel or some other comparable substitute. In the contemporary era, sophisticated electronic devices rely on components that are made from what are called *semiconducting materials*.

[1]The approximate dates for the beginnings of the Stone, Bronze, and Iron ages are 2.5 million BC, 3500 BC, and 1000 BC, respectively.

## 1.2 MATERIALS SCIENCE AND ENGINEERING

Sometimes it is useful to subdivide the discipline of materials science and engineering into *materials science* and *materials engineering* subdisciplines. Strictly speaking, materials science involves investigating the relationships that exist between the structures and properties of materials (i.e., why materials have their properties). In contrast, materials engineering involves, on the basis of these structure–property correlations, designing or engineering the structure of a material to produce a predetermined set of properties. From a functional perspective, the role of a materials scientist is to develop or synthesize new materials, whereas a materials engineer is called upon to create new products or systems using existing materials and/or to develop techniques for processing materials. Most graduates in materials programs are trained to be both materials scientists and materials engineers.

*Structure* is, at this point, a nebulous term that deserves some explanation. In brief, the structure of a material usually relates to the arrangement of its internal components. Structural elements may be classified on the basis of size and in this regard there are several levels:

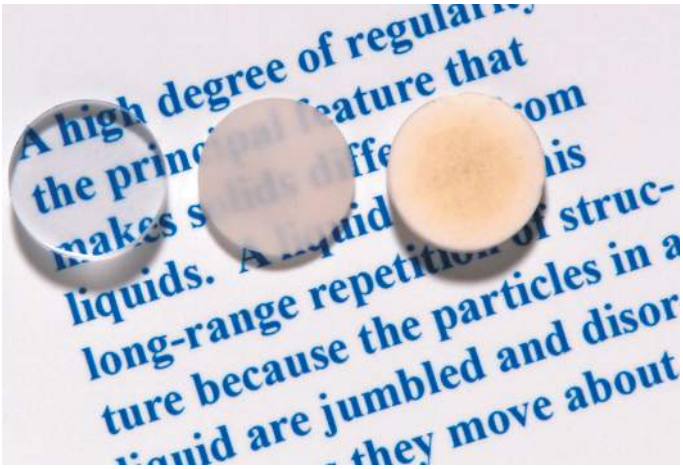
- Subatomic structure—involves electrons within the individual atoms, their energies and interactions with the nuclei.
- Atomic structure—relates to the organization of atoms to yield molecules or crystals.
- Nanostructure—deals with aggregates of atoms that form particles (nanoparticles) that have nanoscale dimensions (less that about 100 nm).
- Microstructure—those structural elements that are subject to direct observation using some type of microscope (structural features having dimensions between 100 nm and several millimeters).
- Macrostructure—structural elements that may be viewed with the naked eye (with scale range between several millimeters and on the order of a meter).

Atomic structure, nanostructure, and microstructure of materials are investigated using microscopic techniques discussed in Section 4.10.

The notion of *property* deserves elaboration. While in service use, all materials are exposed to external stimuli that evoke some types of responses. For example, a specimen subjected to forces experiences deformation, or a polished metal surface reflects light. A property is a material trait in terms of the kind and magnitude of response to a specific imposed stimulus. Generally, definitions of properties are made independent of material shape and size.

Virtually all important properties of solid materials may be grouped into six different categories: mechanical, electrical, thermal, magnetic, optical, and deteriorative. For each, there is a characteristic type of stimulus capable of provoking different responses. These are noted as follows:

- Mechanical properties—relate deformation to an applied load or force; examples include elastic modulus (stiffness), strength, and resistance to fracture.
- Electrical properties—the stimulus is an applied electric field; typical properties include electrical conductivity and dielectric constant.
- Thermal properties—are related to changes in temperature or temperature gradients across a material; examples of thermal behavior include thermal expansion and heat capacity.
- Magnetic properties—the responses of a material to the application of a magnetic field; common magnetic properties include magnetic susceptibility and magnetization.
- Optical properties—the stimulus is electromagnetic or light radiation; index of refraction and reflectivity are representative optical properties.




William D. Callister Jr./ Specimen preparation, P.A. Lessing

**Figure 1.1** Three thin disk specimens of aluminum oxide that have been placed over a printed page in order to demonstrate their differences in light-transmittance characteristics. The disk on the left is *transparent* (i.e., virtually all light that is reflected from the page passes through it), whereas the one in the center is *translucent* (meaning that some of this reflected light is transmitted through the disk). The disk on the right is *opaque*—that is, none of the light passes through it. These differences in optical properties are a consequence of differences in structure of these materials, which have resulted from the way the materials were processed.

- Deteriorative characteristics—relate to the chemical reactivity of materials; for example, corrosion resistance of metals.

The chapters that follow discuss properties that fall within each of these six classifications.

In addition to structure and properties, two other important components are involved in the science and engineering of materials—namely, *processing* and *performance*. With regard to the relationships of these four components, the structure of a material depends on how it is processed. Furthermore, a material's performance is a function of its properties.

We present an example of these processing-structure-properties-performance principles in Figure 1.1, a photograph showing three thin disk specimens placed over some printed matter. It is obvious that the optical properties (i.e., the light transmittance) of each of the three materials are different; the one on the left is transparent (i.e., virtually all of the reflected light from the printed page passes through it), whereas the disks in the center and on the right are, respectively, translucent and opaque. All of these specimens are of the same material, aluminum oxide, but the leftmost one is what we call a *single crystal*—that is, has a high degree of perfection—which gives rise to its transparency. The center one is composed of numerous and very small single crystals that are all connected; the boundaries between these small crystals scatter a portion of the light reflected from the printed page, which makes this material optically translucent. Finally, the specimen on the right is composed not only of many small, interconnected crystals, but also of a large number of very small pores or void spaces. These pores scatter the reflected light to a greater degree than the crystal boundaries and render this material opaque. Thus, the structures of these three specimens are different in terms of crystal boundaries and pores, which affect the optical transmittance properties. Furthermore, each material was produced using a different processing technique. If optical transmittance is an important parameter relative to the ultimate in-service application, the performance of each material will be different.

This interrelationship among processing, structure, properties, and performance of materials may be depicted in linear fashion as in the schematic illustration shown in Figure 1.2. The model represented by this diagram has been called by some the *central paradigm of materials science and engineering* or sometimes just the *materials paradigm*. (The term "paradigm" means a model or set of ideas.) This paradigm, formulated in the 1990s is, in essence, the core of the discipline of materials science and engineering. It describes the protocol for selecting and designing materials for specific and well-defined




**Figure 1.2** The four components of the discipline of materials science and engineering and their interrelationship.

applications, and has had a profound influence on the field of materials.[2] Previous to this time the materials science/engineering approach was to design components and systems using the existing palette of materials. The significance of this new paradigm is reflected in the following quotation: ". . . whenever a material is being created, developed, or produced, the properties or phenomena the material exhibits are of central concern. Experience shows that the properties and phenomena associated with a material are intimately related to its composition and structure at all levels, including which atoms are present and how the atoms are arranged in the material, and that this structure is the result of synthesis and processing."[3]

Throughout this text, we draw attention to the relationships among these four components in terms of the design, production, and utilization of materials.

## 1.3 WHY STUDY MATERIALS SCIENCE AND ENGINEERING?

Why do engineers and scientists study materials? Simply, because things engineers design are made of materials. Many an applied scientist or engineer (e.g., mechanical, civil, chemical, electrical), is at one time or another exposed to a design problem involving materials—for example, a transmission gear, the superstructure for a building, an oil refinery component, or an integrated circuit chip. Of course, materials scientists and engineers are specialists who are totally involved in the investigation and design of materials.

Many times, an engineer has the option of selecting a best material from the thousands available. The final decision is normally based on several criteria. First, the in-service conditions must be characterized, for these dictate the properties required of the material. Only on rare occasions does a material possess the optimum or ideal combination of properties. Thus, it may be necessary to trade one characteristic for another. The classic example involves strength and ductility; normally, a material having a high strength has only a limited ductility. In such cases, a reasonable compromise between two or more properties may be necessary.

A second selection consideration is any deterioration of material properties that may occur during service operation. For example, significant reductions in mechanical strength may result from exposure to elevated temperatures or corrosive environments.

Finally, probably the overriding consideration is that of economics: What will the finished product cost? A material may be found that has the optimum set of properties but is prohibitively expensive. Here again, some compromise is inevitable. The cost of a finished piece also includes any expense incurred during fabrication to produce the desired shape.

The more familiar an engineer or scientist is with the various characteristics and structure–property relationships, as well as the processing techniques of materials, the more proficient and confident he or she will be in making judicious materials choices based on these criteria.

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[2]This paradigm has recently been updated to include the component of material sustainability in the "Modified Paradigm of Materials Science and Engineering," as represented by the following diagram:

Processing → Structure → Properties → Performance → Reuse/Recyclability

[3]"*Materials Science and Engineering for the 1990s*," p. 27, National Academies Press, Washington, DC, 1998.

## CASE STUDY 1.1

### *Liberty Ship Failures*

The following case study illustrates one role that materials scientists and engineers are called upon to assume in the area of materials performance: analyze mechanical failures, determine their causes, and then propose appropriate measures to guard against future incidents.

The failure of many of the World War II Liberty ships[4] is a well-known and dramatic example of the brittle fracture of steel that was thought to be ductile.[5] Some of the early ships experienced structural damage when cracks developed in their decks and hulls. Three of them catastrophically split in half when cracks formed, grew to critical lengths, and then rapidly propagated completely around the ships' girths. Figure 1.3 shows one of the ships that fractured the day after it was launched.

Subsequent investigations concluded one or more of the following factors contributed to each failure:[6]

- When some normally ductile metal alloys are cooled to relatively low temperatures, they become susceptible to brittle fracture—that is, they experience a ductile-to-brittle transition upon cooling through a critical range of temperatures.

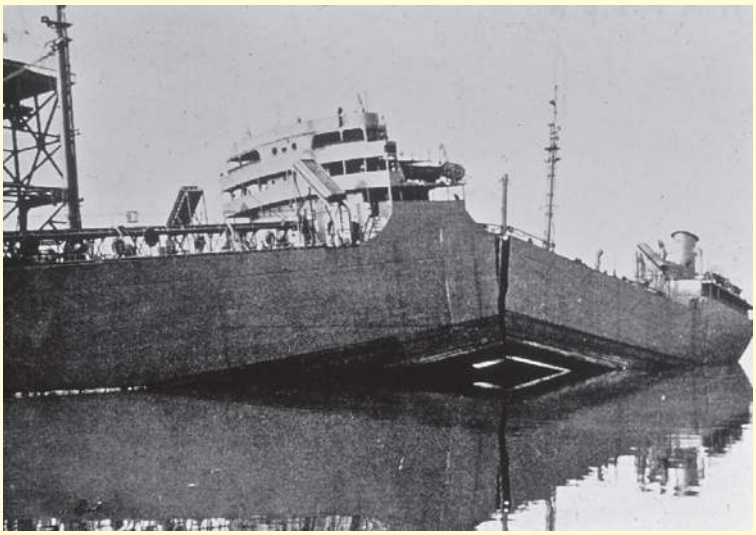


**Figure 1.3** The Liberty ship *S.S. Schenectady,* which, in 1943, failed before leaving the shipyard.
(Reprinted with permission of Earl R. Parker, *Brittle Behavior of Engineering Structures,* National Academy of Sciences, National Research Council, John Wiley & Sons, New York, 1957.)

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[4]During World War II, 2,710 Liberty cargo ships were mass-produced by the United States to supply food and materials to the combatants in Europe.

[5]Ductile metals fail after relatively large degrees of permanent deformation; however, very little if any permanent deformation accompanies the fracture of brittle materials. Brittle fractures can occur very suddenly as cracks spread rapidly; crack propagation is normally much slower in ductile materials, and the eventual fracture takes longer. For these reasons, the ductile mode of fracture is usually preferred. Ductile and brittle fractures are discussed in Sections 8.3 and 8.4.

[6]Sections 8.2 through 8.6 discuss various aspects of failure.